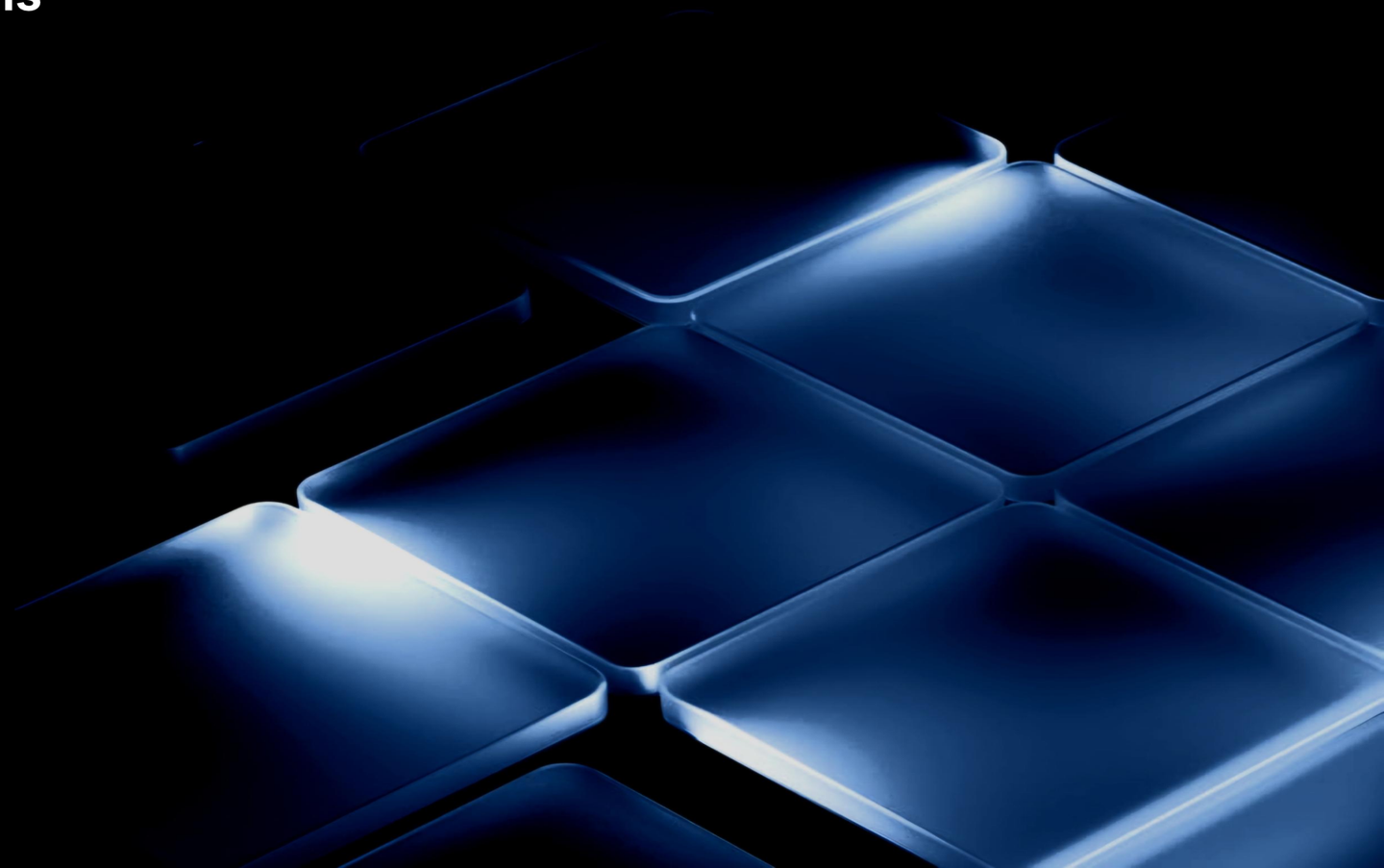
جامعة الكفيل
University of Alkafeel

KALMAN FILTERING

For State Estimation in Intelligent Systems



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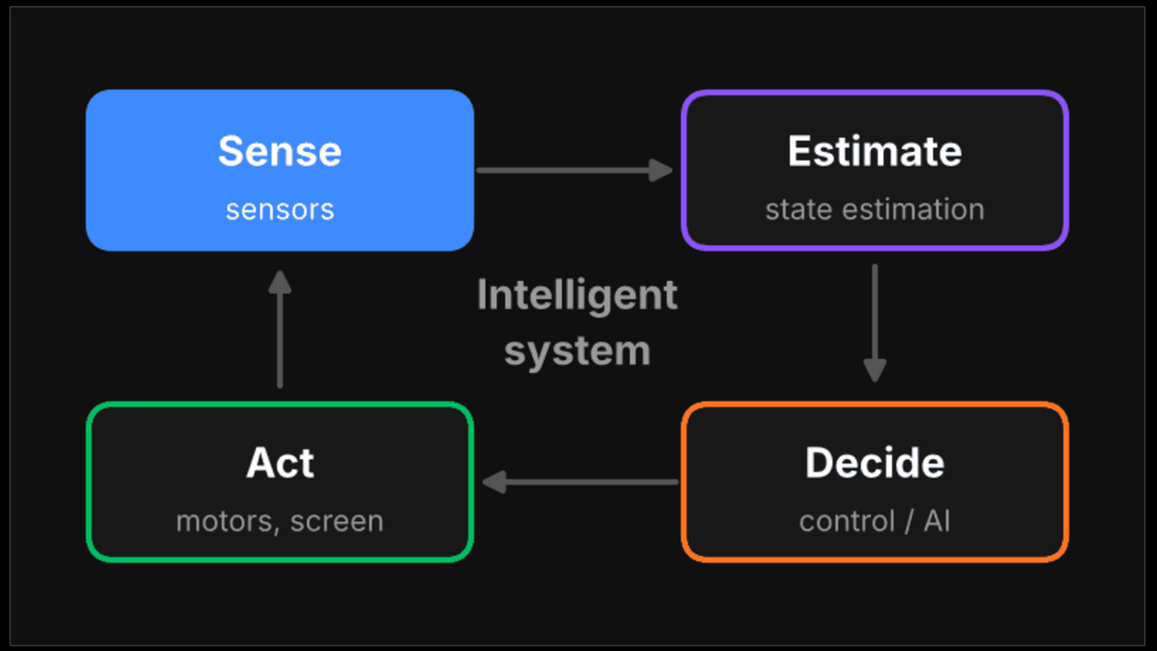
Date: 2026/11/10

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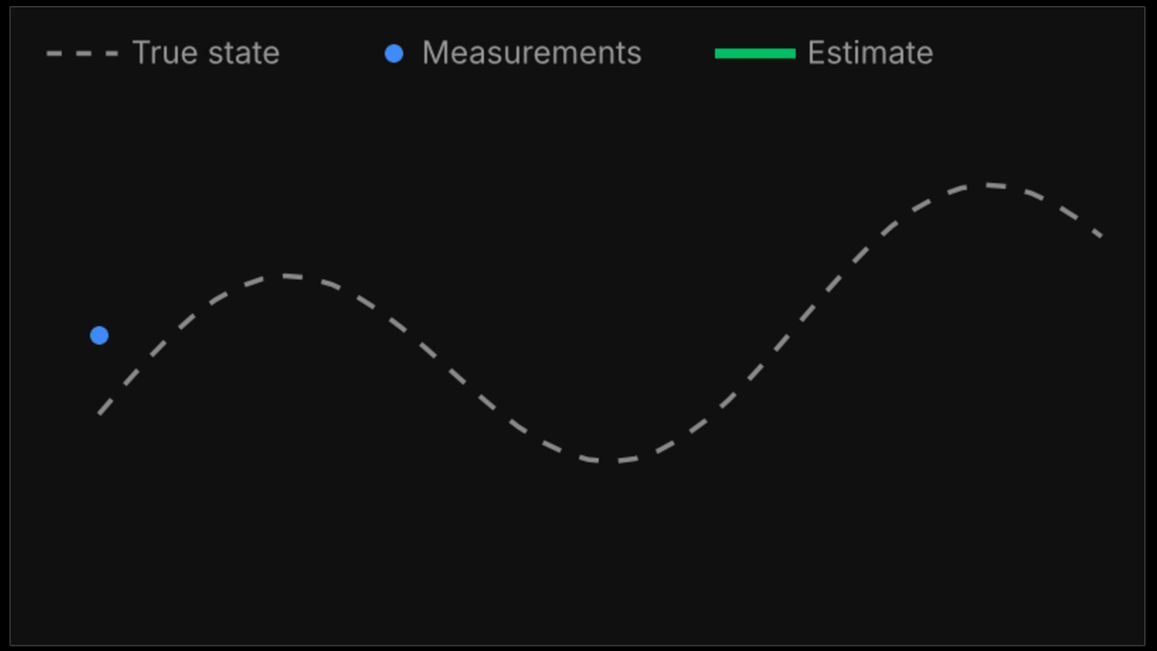
Introduction to Intelligent Systems

- Sense: collect data from the environment
- Estimate: turn noisy data into a reliable state
- Decide: choose what to do
- Act: move, steer, display



What is State Estimation?

- State: what describes the system now (position, speed, angle)
- The true state is hidden
- Sensors give noisy measurements
- Estimation: measurements + a model → the best guess of the state



Why Do We Need State Estimation?

- Noisy sensors: readings fluctuate even at rest
- Hidden states: some quantities are not measured directly
- Lag and dropouts: signals arrive late or disappear
- Control: decisions need reliable values

Noise in Measurements

- Measurement = truth + noise

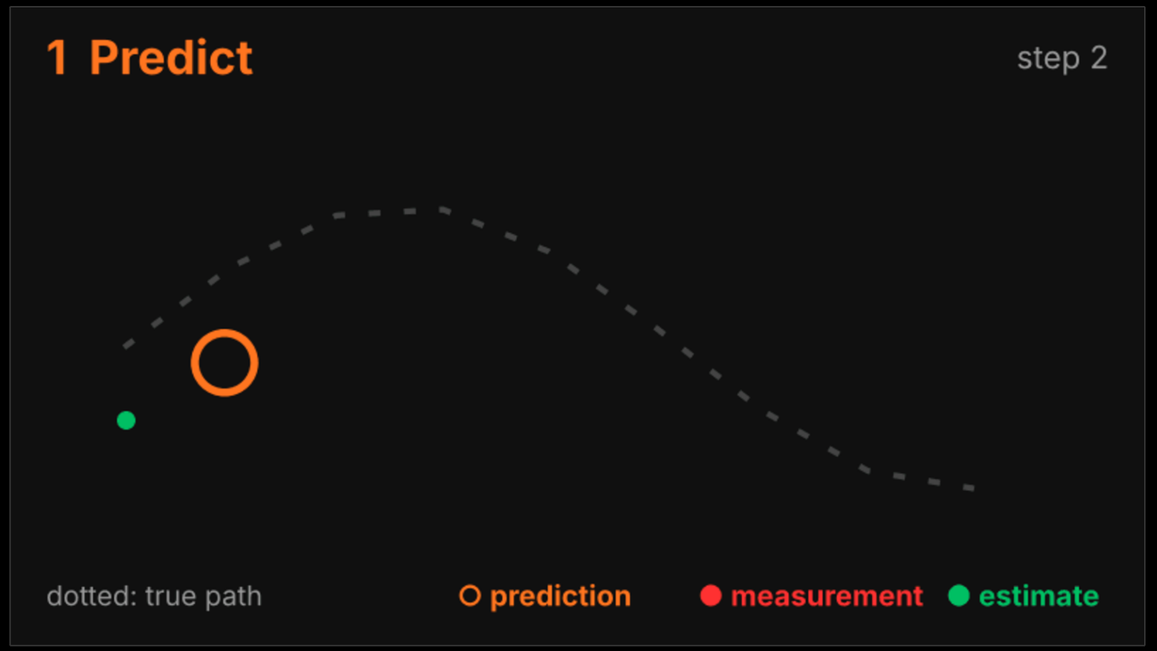
$$z = x + v$$

- Noise is random and bell-shaped (Gaussian)
- Its spread is the variance R



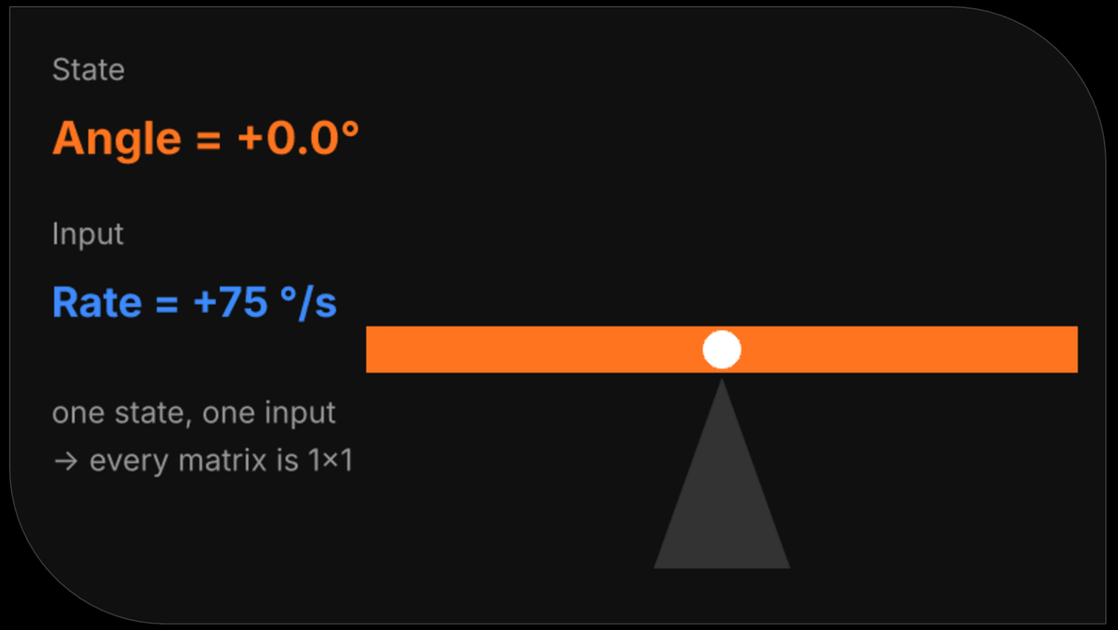
Introduction to Kalman Filtering

- Recursive estimator for noisy measurements
- Predict with a model, update with a measurement
- Optimal for linear systems with Gaussian noise
- Keeps only the previous estimate
- Kálmán, 1960; used in Apollo navigation



State-Space Representation

SYMBOL	MEANING	OUR 1D CASE
x	State: what we want to know	Angle
u	Input: known motion	Rate
z	Measurement: what the sensor reports	Measured angle
A	How the state evolves	1
B	How the input changes the state	dt
H	How the state maps to the measurement	1



System Model and Measurement Model

System model

General $\mathbf{x}_k = \mathrm{A}\,\mathbf{x}_{k-1} + \mathrm{B}\,\mathbf{u}_k + \mathbf{w}_k$

Our 1D case $\mathrm{Angle}_k = \mathrm{Angle}_{k-1} + (\mathrm{Rate}_k + w_k) \times dt$

w: process noise, variance Q

Measurement model

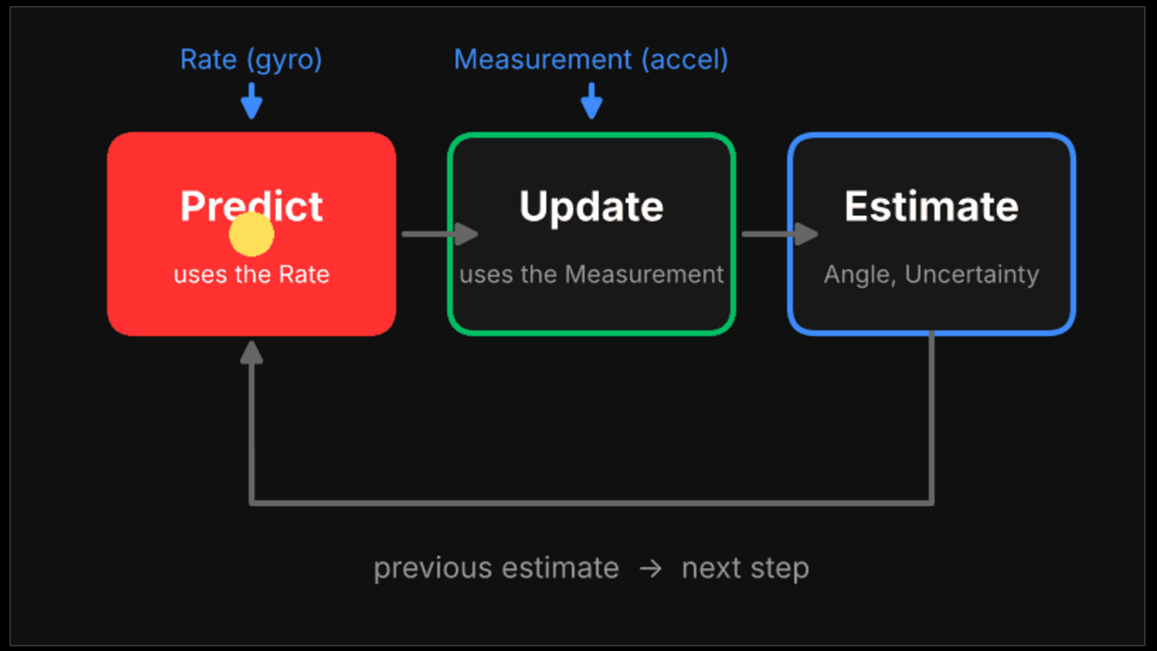
General $\mathbf{z}_k = \mathrm{H}\,\mathbf{x}_k + \mathbf{v}_k$

Our 1D case $\mathrm{Measurement}_k = \mathrm{Angle}_k + v_k$

v: measurement noise, variance R

Basic Kalman Filter Architecture

- Start: initial Angle and Uncertainty
- Predict: with the model
- Update: with the measurement
- Repeat for every new sample



Prediction Step

GENERAL

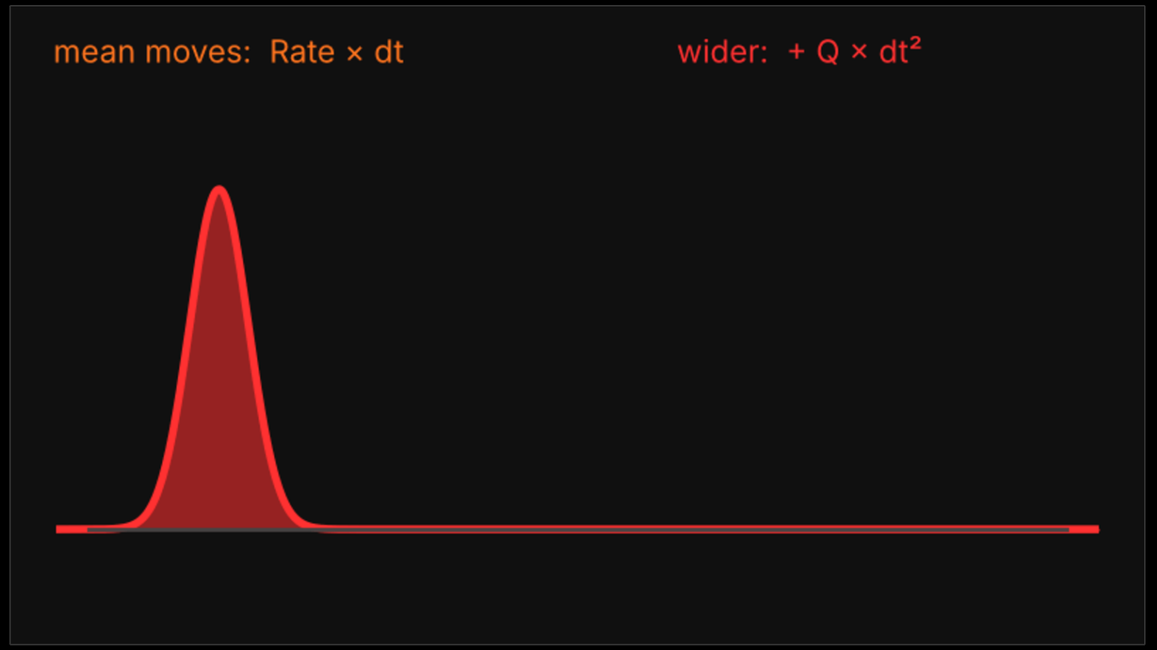
$$x_{pred} = A\,x_{prev} + B\,u$$

$$P_{pred} = A\,P_{prev}\,A^{T} + Q$$

OUR 1D CASE

$$Angle_{pred} = Angle_{prev} + Rate \times dt$$

$$Uncertainty_{pred} = Uncertainty_{prev} + Q \times dt^{2}$$



- The angle moves, the uncertainty grows.

Measurement Update Step

GENERAL

$$x_{est} = x_{pred} + K\,(z - H\,x_{pred})$$

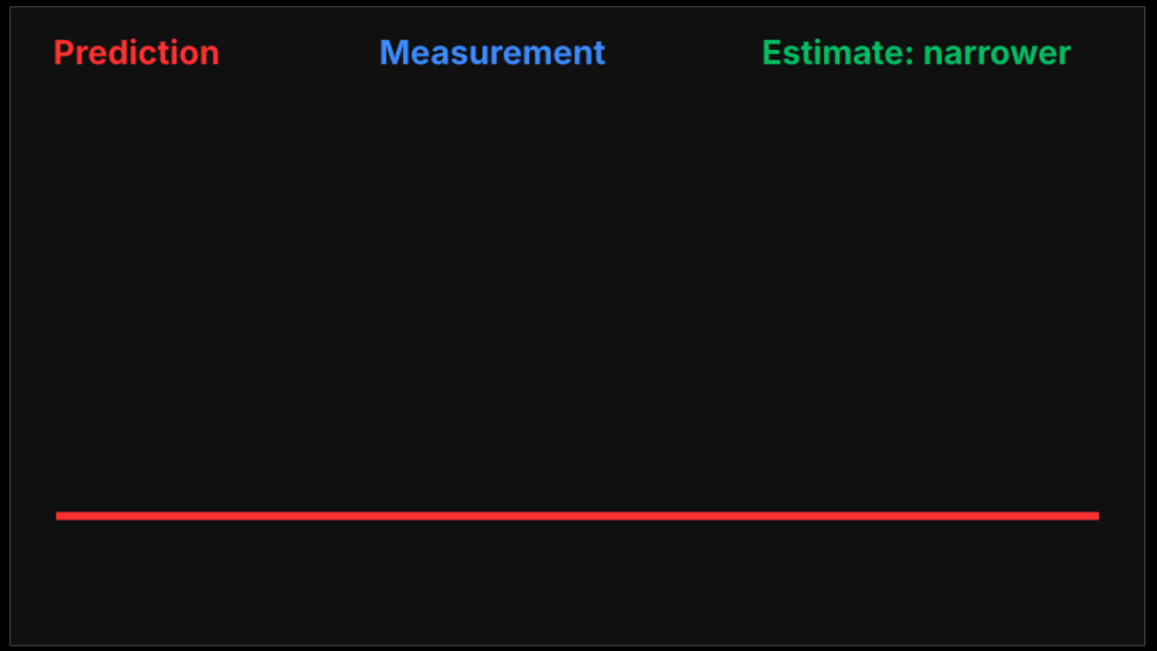
$$P_{est} = (I - K\,H)\,P_{pred}$$

OUR 1D CASE

$$Angle_{est} = Angle_{pred} + K \times (Measurement - Angle_{pred})$$

$$Uncertainty_{est} = (1 - K) \times Uncertainty_{pred}$$

- $Residual = Measurement - Angle_{pred}$



Kalman Gain

GENERAL

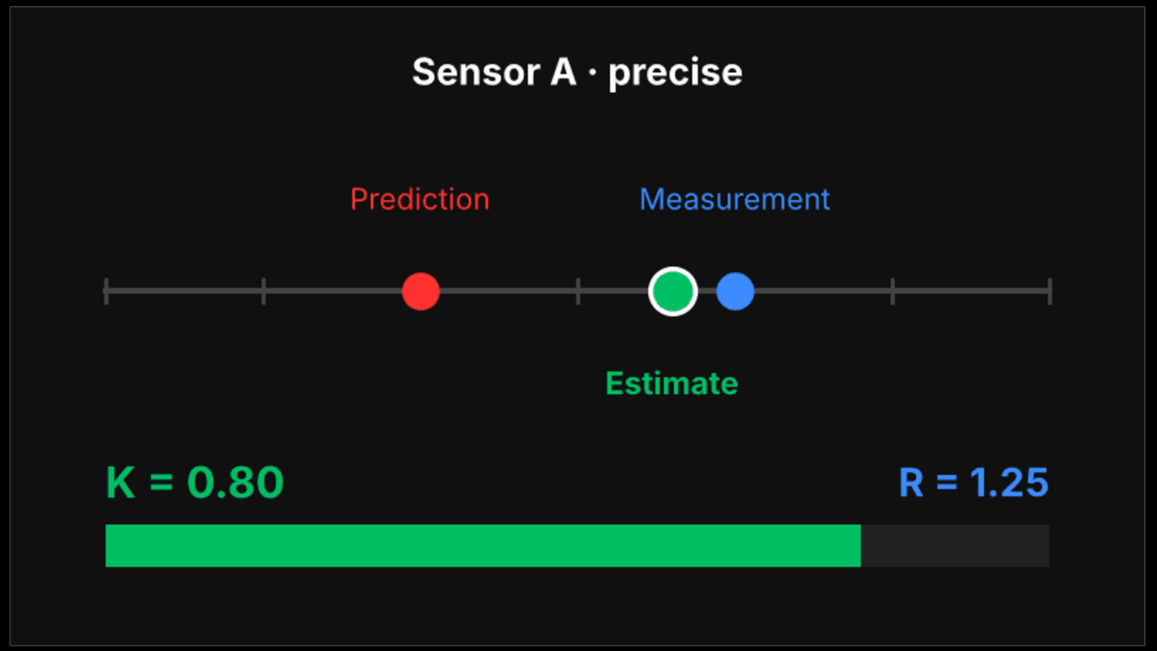
$$K = P_{pred} H^T (H P_{pred} H^T + R)^{-1}$$

OUR 1D CASE

$$K = Uncertainty_{pred} / (Uncertainty_{pred} + R)$$

$K \to 0$
trust the prediction

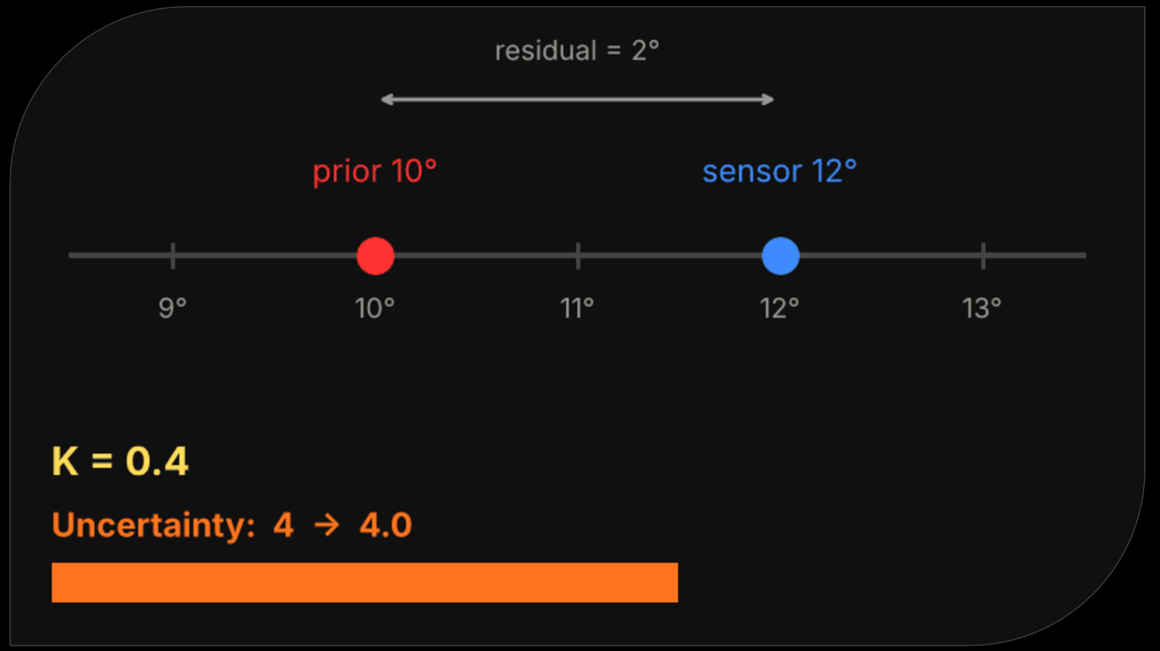
$K \to 1$
trust the measurement



Worked Example

Prediction: 10°, uncertainty 4
Sensor: 12°, R = 6

1. $K = 4 / (4 + 6) = 0.4$
2. Residual = 12 − 10 = 2°
3. $\text{Angle}_{est} = 10 + 0.4 \times 2 = 10.8°$
4. $\text{Uncertainty}_{est} = (1 - 0.4) \times 4 = 2.4$



Practical Implementation Using MPU6050

- Accelerometer: tilt from gravity, but noisy
- Gyroscope: smooth rate, but drifts
- Kalman filter: gyro predicts, accelerometer corrects

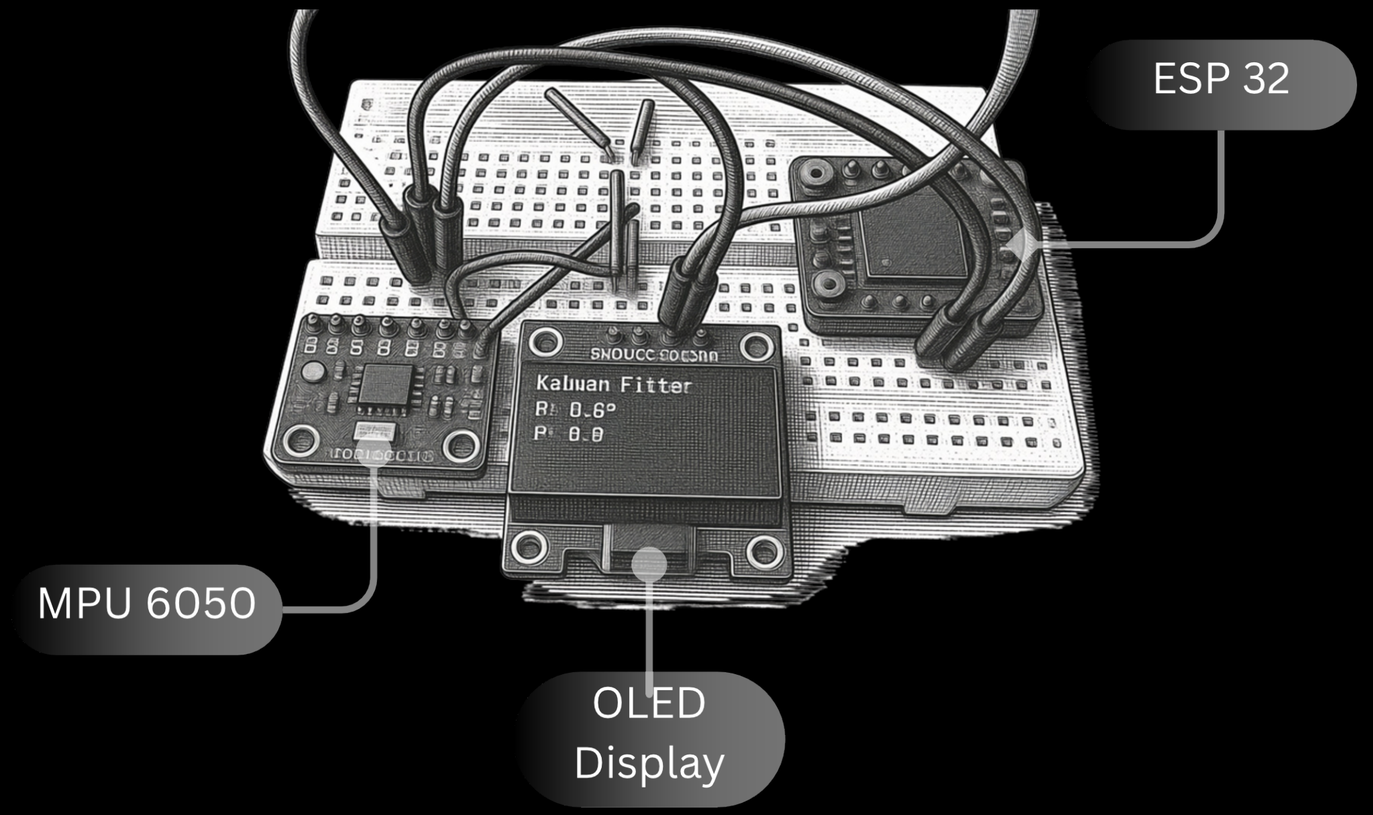


Practical Implementation

The MPU6050 combines an Accelerometer and a Gyroscope. Their readings feed the ESP32-C3, where the Kalman filter estimates orientation before the result reaches the OLED.

Accelerometer: acceleration on X, Y, Z; tilt from gravity.

Gyroscope: rotation rate; angle change from rate × time.



MPU6050 --> ESP32-C3 (Kalman Filter) --> OLED

From Sensor Readings to Filter Inputs

Gyroscope → Rate

$$\text{Rate} = \text{Gyro}_{\text{Rad}} \times (180/\pi) - \text{Bias}$$

$$\text{Bias} = (\Sigma\ \text{Gyro}_{\text{Readings}}) / N$$

N readings with the sensor still

Accelerometer → Measurement

$$\text{Roll} = \text{atan2}(Ay, Az) \times (180/\pi)$$

$$\text{Pitch} = \text{atan2}(-Ax, \sqrt{(Ay^2 + Az^2)}) \times (180/\pi)$$

Ax, Ay, Az: readings on the X, Y, Z axes

From Equations to Code

01
Calibrate
Average 1,000 still gyro readings: Bias

02
Read
Gyro rate and Ax, Ay, Az

03
Measure
Raw Roll and Pitch from the accelerometer

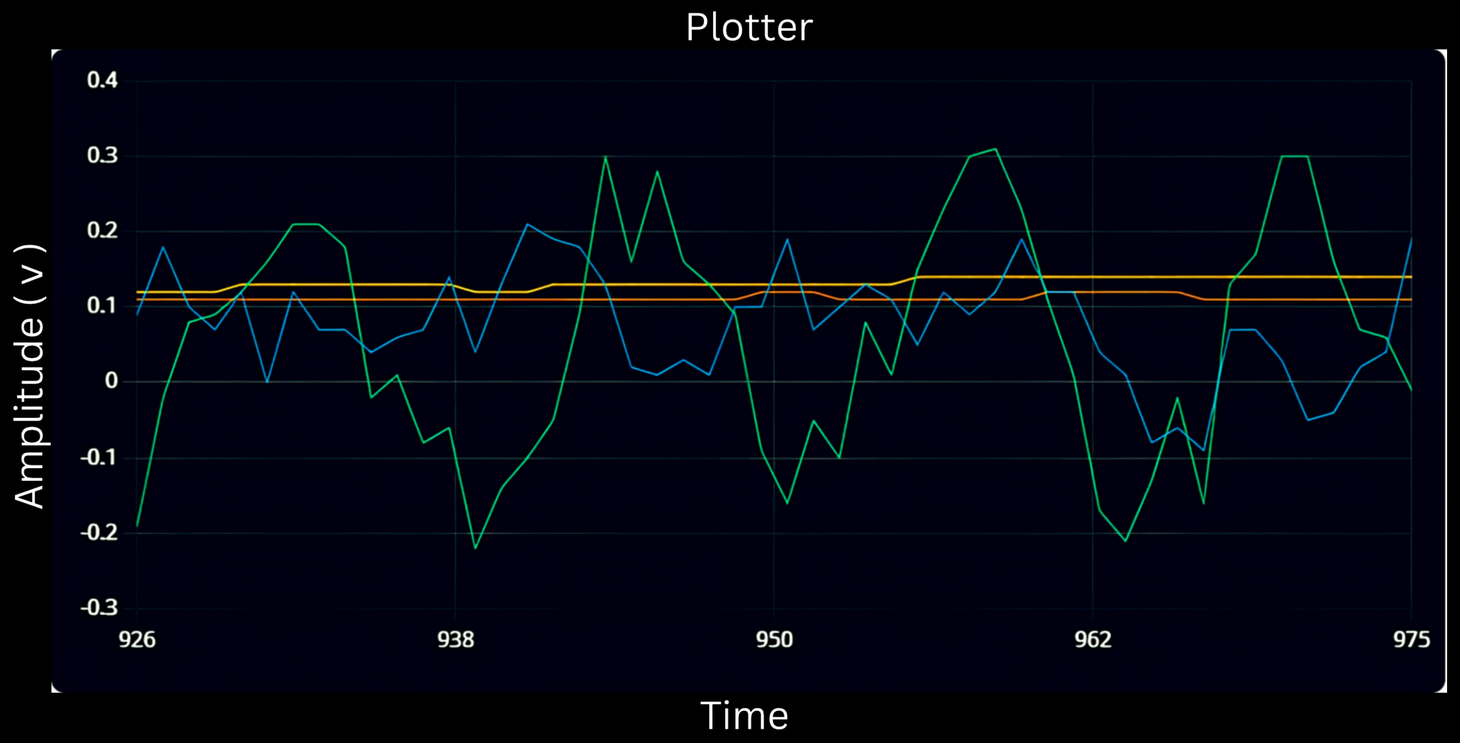
04
Filter
kalman1D(): predict, then update

05
Display
Filtered Roll and Pitch on the OLED

Raw vs filtered signal

- **Raw Roll**
  Roll: Y, Z · sideways tilt
- **Kalman Roll**
- **Raw Pitch**
  Pitch: X · forward / backward tilt
- **Kalman Pitch**

Kalman Roll & Pitch are what matter: the filtered result the application uses.



Raw: rapid fluctuations even when the motion is small, partly noise.

Filtered: a smoother, more stable estimate, shown live on the OLED.

Judging the filter

The trade-off

- Goal: cut the noise, keep the real changes.
- Too slow: the filter lags behind the motion.
- Too smooth: real changes get hidden.

What we saw

- At rest, filtered readings were steadier than raw ones.
- To quantify: standard deviation at rest, or error against a reference angle.



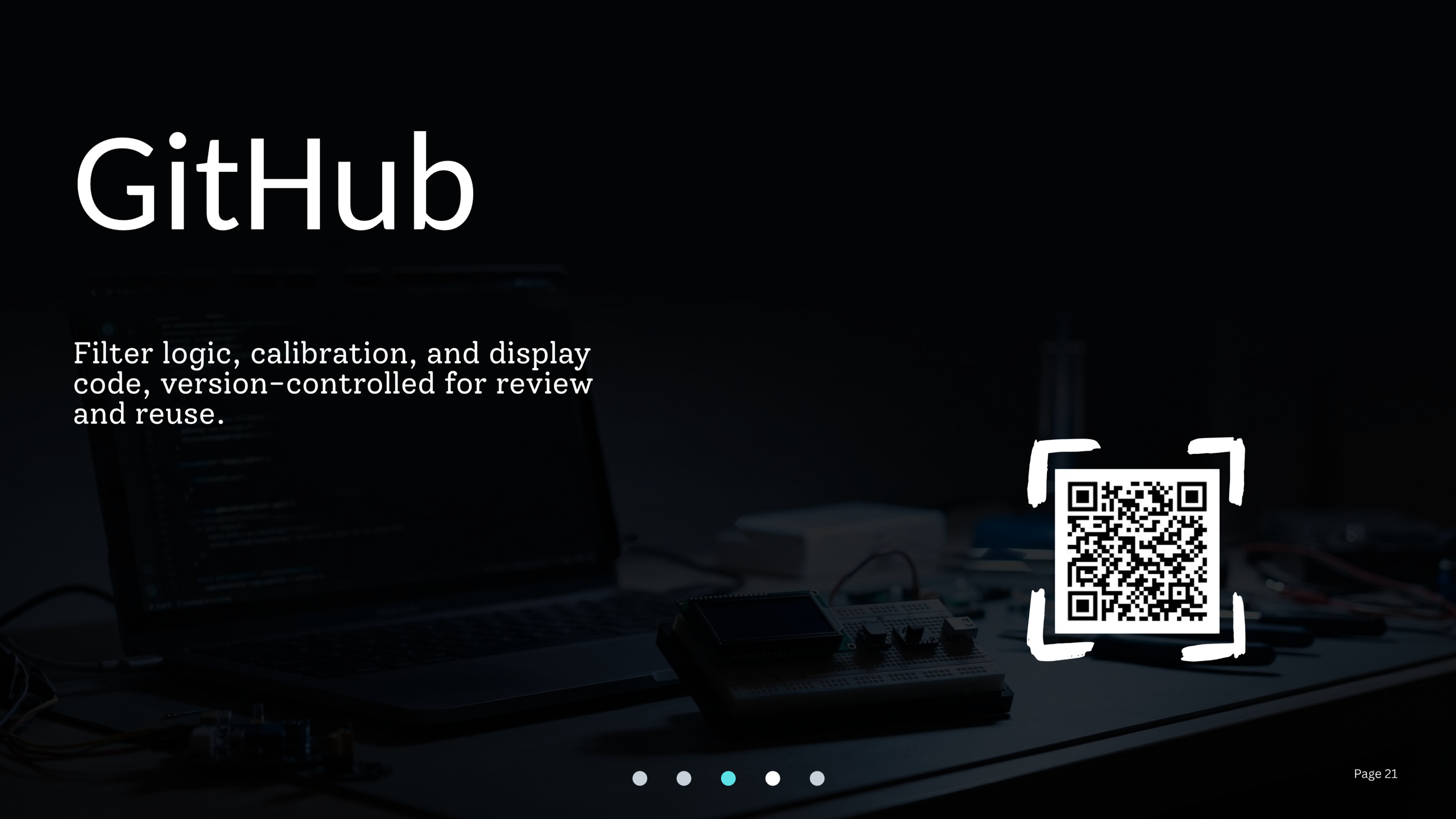
Raw: noisy

The goal: balance

Over-smoothed: laggy

GitHub

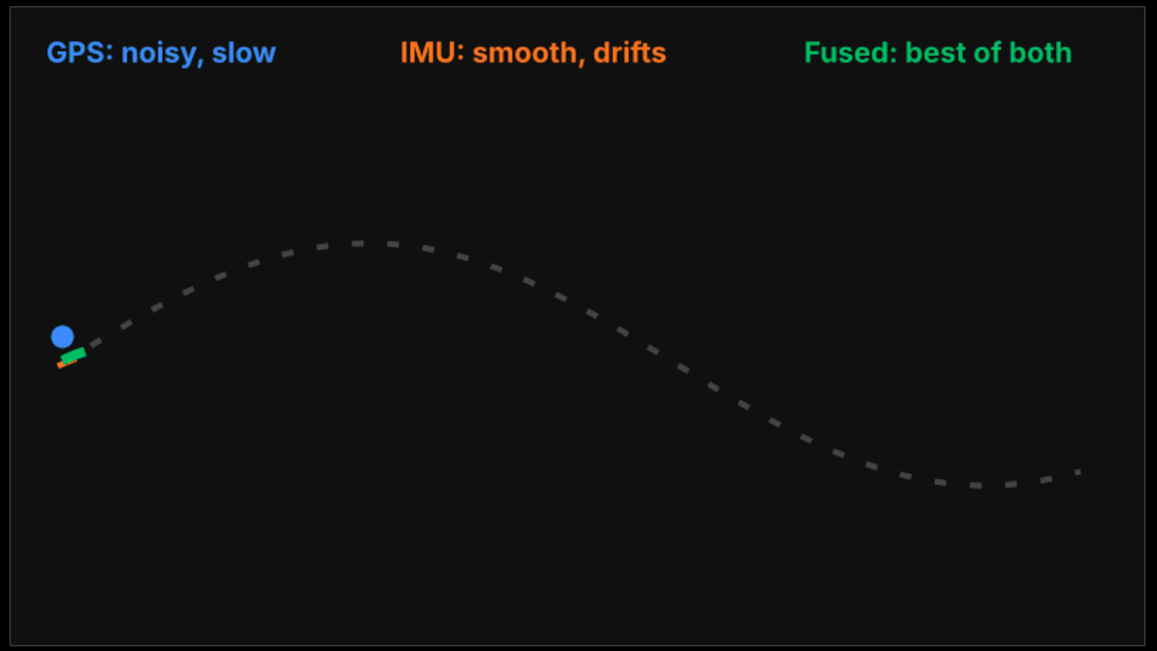
Filter logic, calibration, and display code, version-controlled for review and reuse.



Sensor Fusion

No single sensor is complete. Fusion combines sensors that fail in different ways.

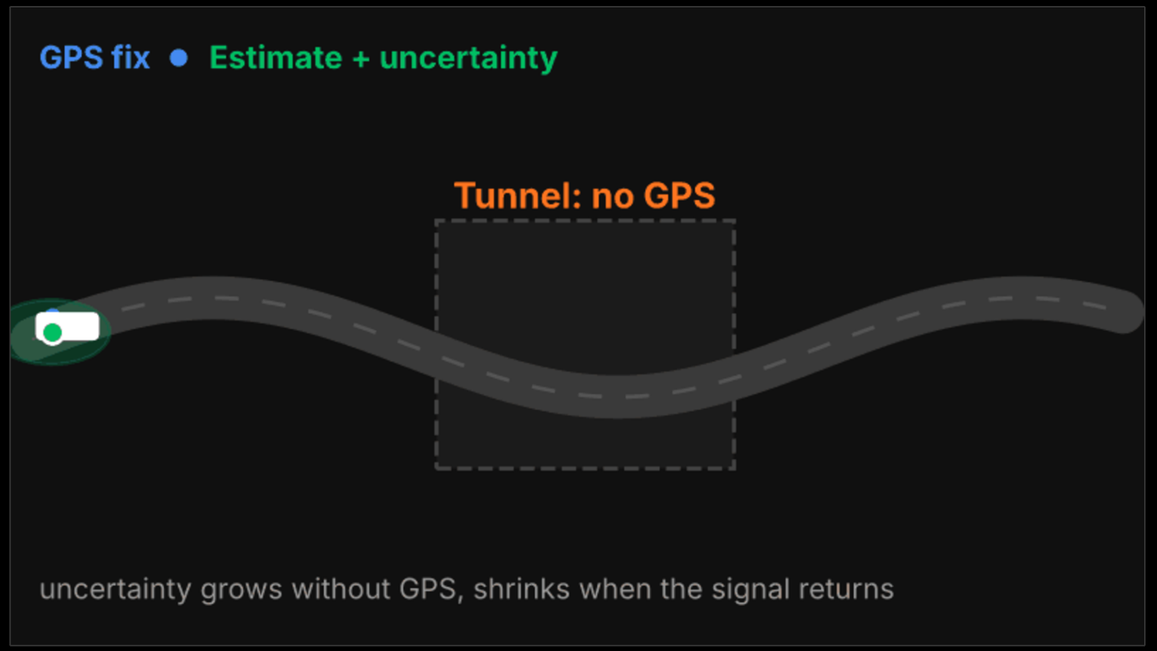
- GPS: absolute position, but noisy and slow.
- Inertial sensor (IMU): fast and smooth, but it drifts.
- Fused: the IMU predicts, GPS corrects, K balances them.



GPS

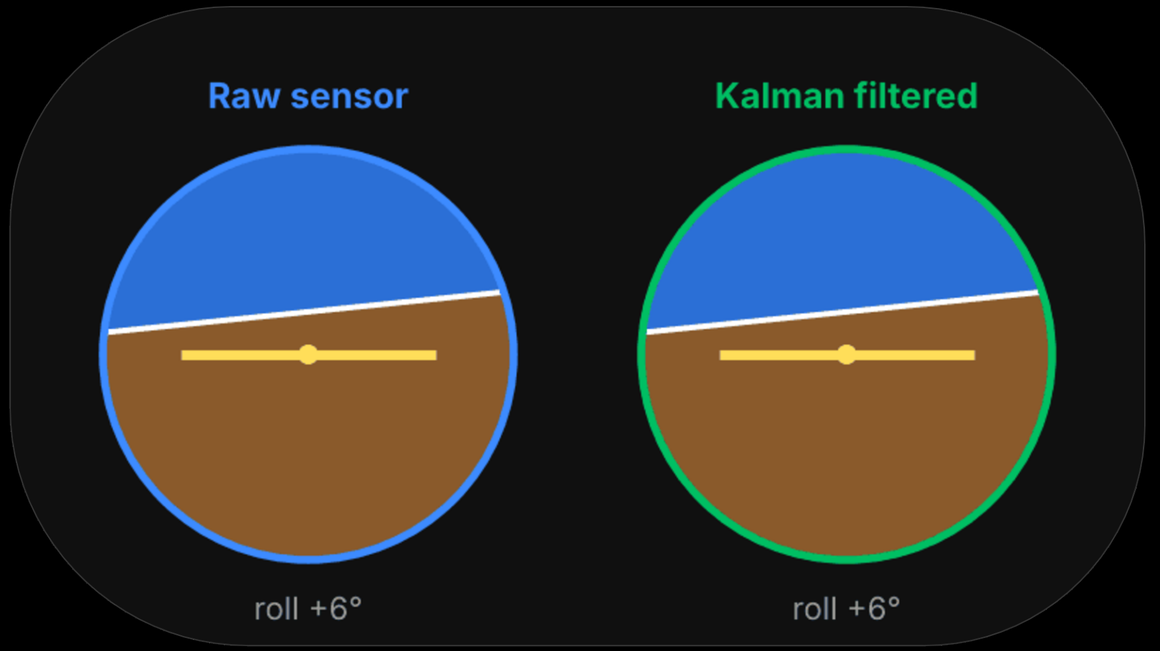
Satellite signals lag or drop between buildings, causing false position jumps.

- Predict: last position plus speed and direction from motion sensors, even with no signal. Uncertainty grows.
- Correct: each GPS fix is compared with the prediction; K balances them. Weak signal (large R): trust the prediction.
- Result: a continuous, stable path with no false jumps.



Drones and Aircraft

- Drones and robots: air currents and sensor noise spoil the attitude; gyroscope and accelerometer are fused at very high speed.
- Commercial aircraft: the autopilot fuses altimeter, airspeed and inertial data; each reading is weighed by its reliability.
- Result: stable flight and steady guidance through turbulence.



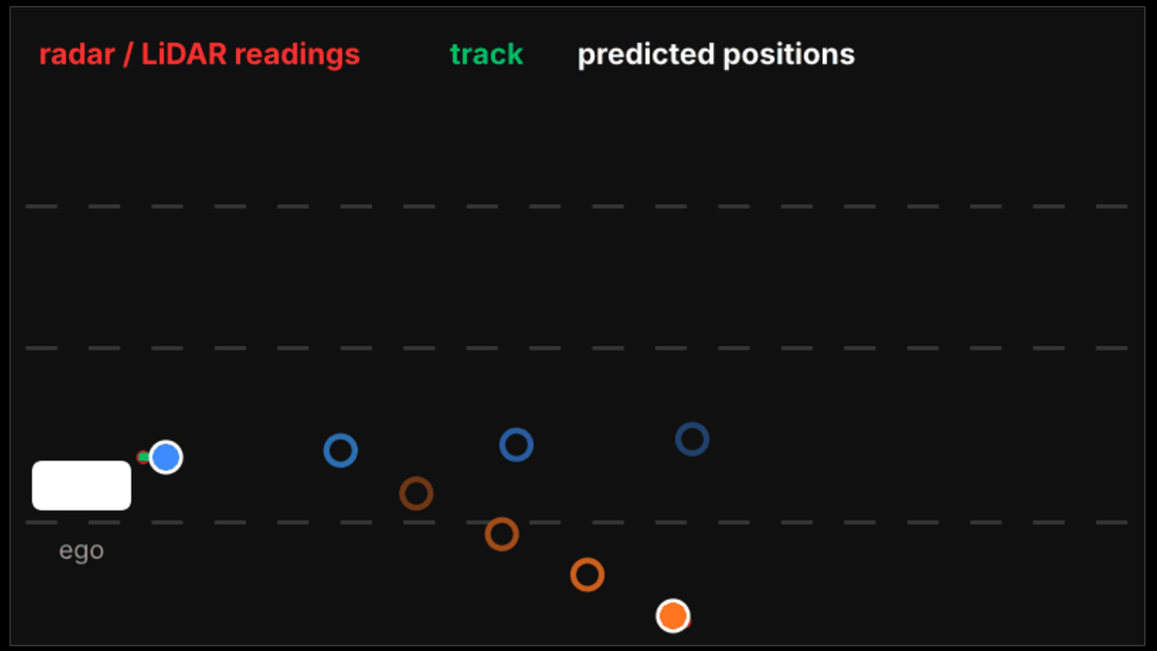
BALANCE LOOP

Gyro predicts → Accel corrects → Tilt estimated → Motors adjust

Self-Driving Cars

The car must know its position and its surroundings precisely, yet every sensor is incomplete or noisy.

- Locating the car: predict from speed, steering angle and motion sensors; correct with GPS or LiDAR matched to the map. In tunnels, prediction carries on.
- Tracking surroundings: predict where vehicles and pedestrians will be; correct with radar, LiDAR and camera. Tracking survives brief dropouts.



Limitations of the Standard Kalman Filter

Linear models

- Assumes the system is linear.
- Turning cars and rotating bodies are nonlinear.

Gaussian noise

- Optimal only for Gaussian noise.
- Outliers and heavy-tailed errors degrade it.

Needs tuning

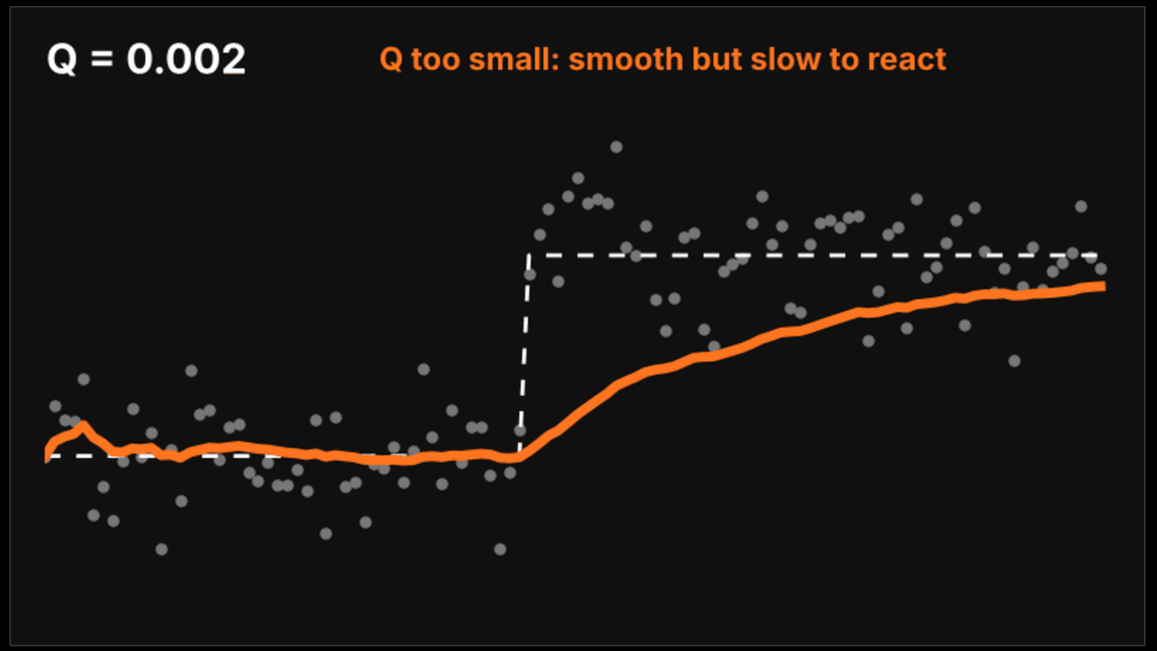
- Q, R and the model must be right.
- Wrong values give lag, noise or divergence.

Imperfect sensors

- Drift, bias and vibration remain.
- Calibration needs the sensor to stay still.

Process and Measurement Noise Tuning

- R, measurement noise: measure it: the variance of the sensor readings at rest.
- Q, process noise: tune it on real data: start small, increase until the filter follows real motion.



EFFECT

Q up or R down: faster response, noisier output.

Q down or R up: smoother output, slower response.

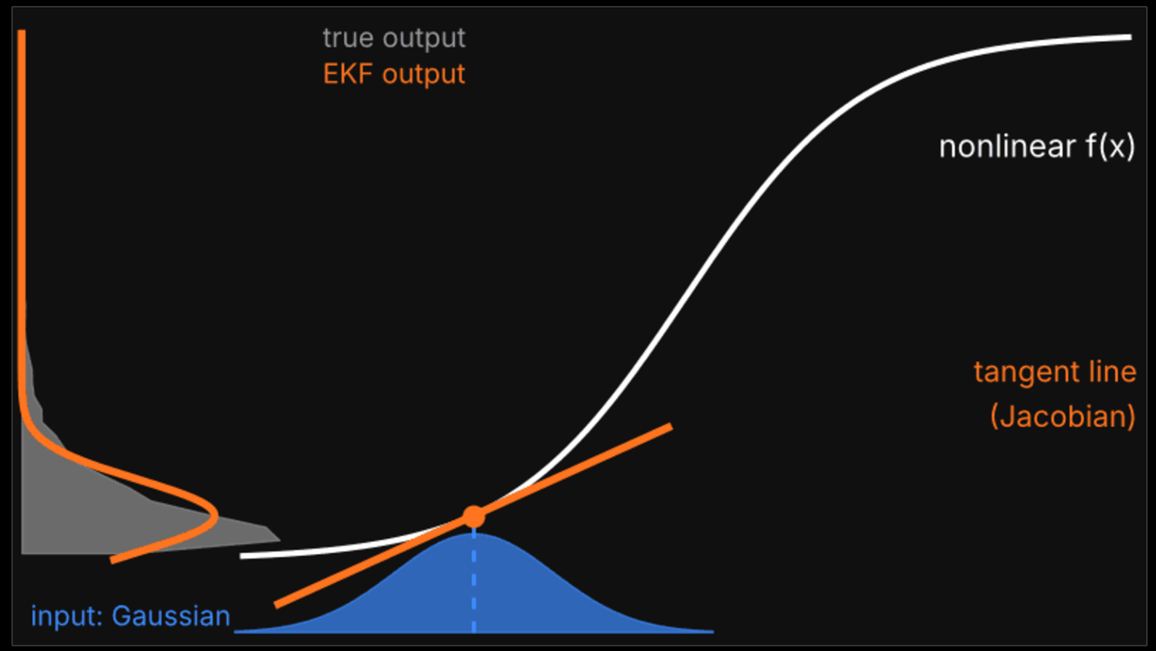
Extended Kalman Filter (EKF)

NONLINEAR MODEL

$x_k = f(x_{k-1}, u_k) \quad z_k = h(x_k)$

$F = \partial f/\partial x \quad H = \partial h/\partial x$

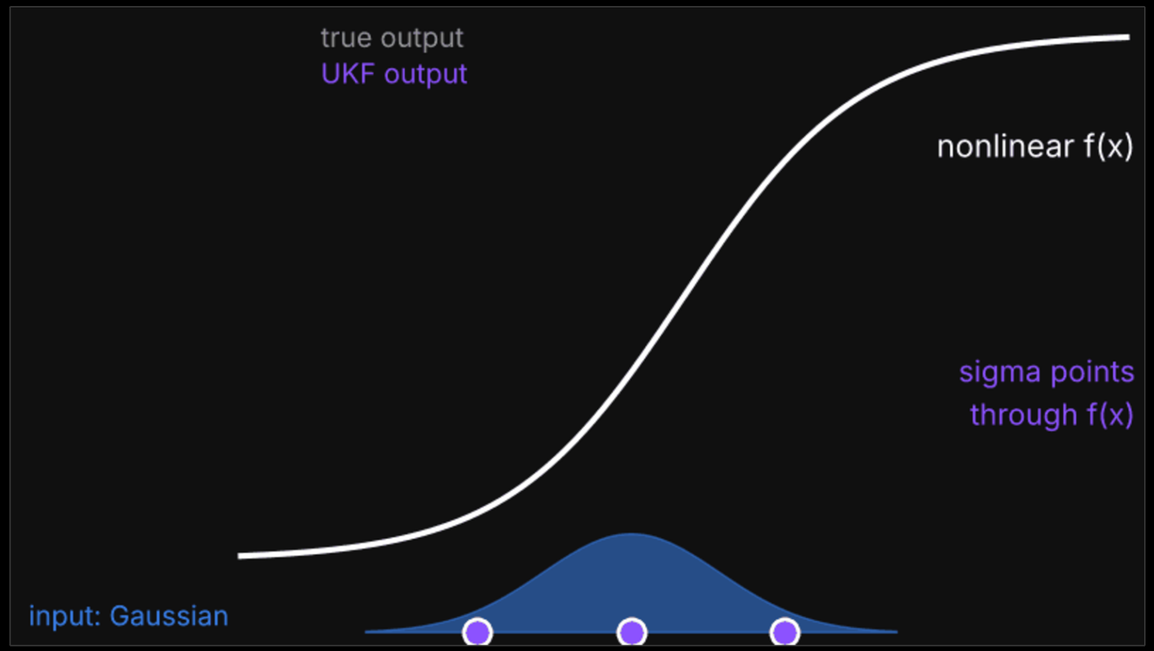
- Linearize f and h around the current estimate (Jacobians F and H).
- Then run the same predict and update steps.
- Used in attitude, GPS/INS and robot or car navigation.
- Limit: only a first-order approximation.



Unscented Kalman Filter (UKF)

Approximate the distribution, not the function.

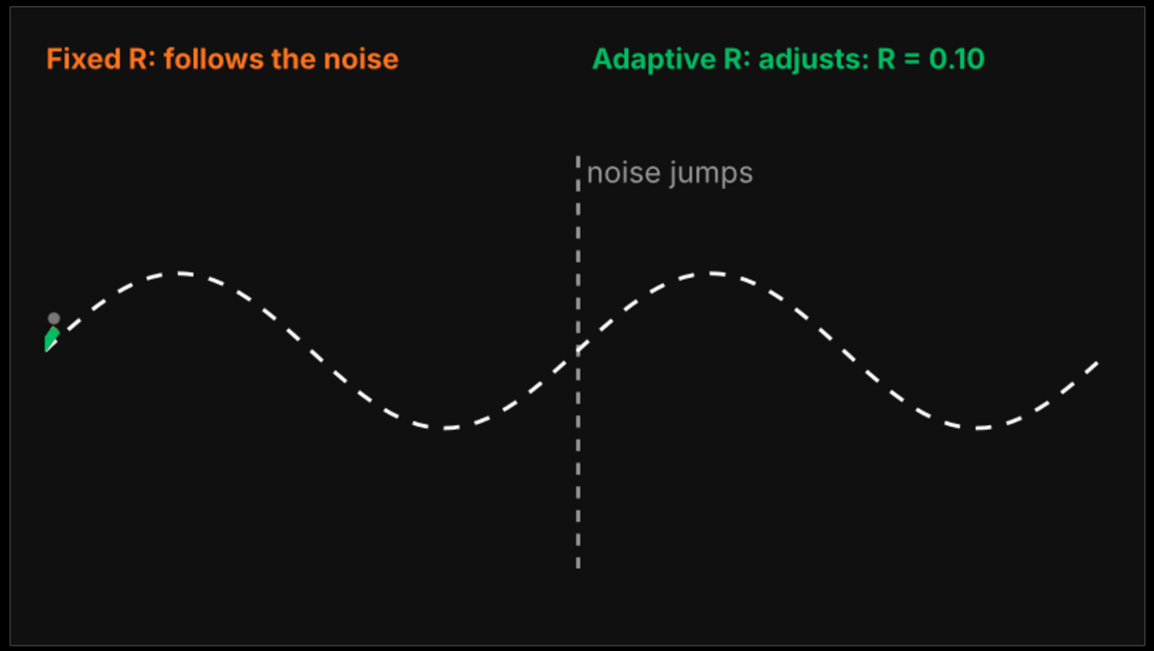
- Pick a few sigma points around the estimate.
- Pass each one through the true nonlinear function.
- Rebuild the mean and covariance from the results.
- No Jacobians; better than the EKF for strong nonlinearity; slightly more computation.



Adaptive Kalman Filtering

Real noise changes. A fixed Q and R cannot follow it.

- Monitor the innovation: the gap between measurement and prediction.
- Larger than expected: increase R (or Q). Smaller: decrease it.
- Variants: innovation-based and multiple-model adaptive filters.
- Cost: extra complexity; the adaptation itself needs tuning.



Choosing the Right Filter

FILTER	MODEL	BEST FOR	COST
Kalman	Linear, Gaussian noise	IMU angle, simple tracking	Lowest
EKF	Mildly nonlinear (Jacobians)	GPS/INS, attitude, robots	Low
UKF	Strongly nonlinear (sigma points)	Complex attitude and tracking	Medium
Adaptive	Changing noise levels	Variable environments	Medium to high

Conclusion

Noisy sensors in, a dependable estimate out.

Estimate

State estimation infers the hidden state from noisy measurements.

Predict + update

The Kalman filter predicts with a model, corrects with the sensor, weighted by K.

Proven in practice

MPU6050 demo: filtered Roll and Pitch are steadier than the raw ones.

Beyond the basics

Tuning, EKF, UKF and adaptive filters extend it to real systems.

Thank you. Questions?

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